

```

const handleDataChange = (newData) => {
  setData(newData);
  saveSpreadsheetData(newData).catch(err => {
    console.error('Error saving data:', err);
  });
};

if (loading) {
  return <div className="flex justify-center items-center h-screen">Loading...</div>;
}

return (
  <CopilotKit
    apiKey={process.env.NEXT_PUBLIC_OPENAI_API_KEY}
    className="min-h-screen"
  >
    <div className="container mx-auto p-4">
      <header className="mb-8">
        <h1 className="text-3xl font-bold">RainbowBeast Copilot Spreadsheet</h1>
        <p className="text-gray-600">
          An intelligent spreadsheet with AI assistance
        </p>
      </header>

      <div className="grid grid-cols-1 lg:grid-cols-3 gap-6">
        <div className="lg:col-span-2 border rounded-lg shadow">
          <div className="bg-gray-50 p-4 rounded-t-lg border-b">
            <h2 className="text-xl font-semibold">Spreadsheet</h2>
          </div>
          <div className="p-4 overflow-x-auto">
            <Spreadsheet data={data} onChange={handleDataChange} />
          </div>
        </div>

        <div className="border rounded-lg shadow">
          <div className="bg-gray-50 p-4 rounded-t-lg border-b">
            <h2 className="text-xl font-semibold">Copilot Assistant</h2>
          </div>
          <div className="p-4 flex flex-col h-[500px]">
            <div className="flex-grow mb-4 overflow-y-auto">
              <CopilotSidebar />
            </div>
            <div className="border rounded">
              <CopilotTextarea
                value={message}
                onChange={(e) => setMessage(e.target.value)}
                placeholder="Ask the AI about your spreadsheet data..."
                className="w-full p-2 min-h-[100px]"
                copilotContext={{
                  spreadsheetData: data
                }}
              />
            </div>
          </div>
        </div>
      </div>
    </CopilotKit>
  );
}

// API ROUTE FOR SPREADSHEET DATA
const apiRoute = `// web/app/api/spreadsheet/route.ts
import { NextResponse } from 'next/server';

```